# Gayatri Vidya Parishad Institute of Higher Learning and Research

**(Deemed to be University under Distinct Category)**

**Kommadi, Madhurawada, Visakhapatnam – 530 048, Andhra Pradesh**

## CALCULUS AND LINEAR ALGEBRA

## LECTURE SCHEDULE

**(Common to ALL Branches)**

**ACADEMIC YEAR: 2026-2027**

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| Pre-requisite Course/Topic Names: | **1. Basics of Limits and Continuity**<br>**2. Basics of Differential and Integral calculus**<br>**3. Elementary Matrix Algebra** |
| **Lecture No** | **Topic** |
| 1 | Orientation of the course |
| **UNIT-I : SEQUENCES, SERIES AND MEAN VALUE THEOREMS** | |
| 2 | Introduction: Definition of a Sequence, convergence of a sequence |
| 3 | Infinite series, Convergence/ divergence of an Infinite series |
| 4 | Convergence of Geometric, Harmonic p-series preliminary test for divergence |
| | **Tutorial -I** |
| 5 | Comparison Test |
| 6 | D'Alembert's Ratio Test |
| 7 | Cauchy's nth root test |
| | **Tutorial -II** |

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| **8** | Rolle's theorem, Lagrange Mean Value theorem |
| **9** | Cauchy's Mean value theorem |
| **10** | Taylor's theorem; Taylor's series expansion of f(x); Maclaurin's Series expansion of f(x) |
| **11** | Summarization of Unit |
| | **Tutorial -III** (Short assessment) |
| **UNIT-II: PARTIAL DIFFERENTIATION** | |
| **12** | Introduction to partial differentiation |
| **13** | Total derivative & Chain rule |
| **14** | Jacobian, Properties |
| | **Tutorial -IV** |
| **15** | Functional dependence - Jacobian |
| **16** | Taylor's series for function of two variables |
| **17** | Maxima and Minima of functions of two variables without constraints |
| | **Tutorial -V** |
| **18** | Lagrange's method of undetermined multipliers |
| **19** | Lagrange's method of undetermined multipliers |
| **20** | Summarization of Unit |
| | **Tutorial -VI** (Short assessment) |
| **UNIT-III: MULTIPLE INTEGRALS** | |
| **21** | Evaluation of Double integrals |
| **22** | Changing the order of Integration/ |
| **23** | Change of variables in double integrals: Cartesian to Polar |
| | **Tutorial -VII** |
| **24-25** | Area enclosed by a closed curve |
| **26, 27** | Evaluation of Triple integrals |
| | **Tutorial -VIII** |
| **28** | Change of variables in Triple integrals: Cartesian to Spherical, Cartesian to cylindrical |
| **29** | Problems |

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| **30** | Summarization of Unit |
| | **Tutorial -IX (**Short assessment) |
| | **UNIT-IV: SOLVING SYSTEMS OF LINEAR EQUATIONS** |
| **31** | Rank of a Matrix – Definition, Rank of a matrix by echelon form |
| **32** | Rank of a matrix by normal form |
| **33-34** | Consistency of Linear system of Equations: Homogeneous, Non- Homogeneous |
| | **Tutorial -X** |
| **35-36** | Eigen Values & Eigen Vectors |
| **37** | Properties of Eigen values and Eigen Vectors |
| | **Tutorial -XI** |
| **38** | Summarization of Unit |
| | **Tutorial -XII (**Short assessment) |
| | **UNIT V: CAYLEY HAMILTON THEOREM AND QUADRATIC FORMS** |
| **39** | Cayley-Hamilton theorem |
| **40** | Application – Finding Inverse and Power of a matrix |
| **41** | Reduction to Diagonal form |
| | **Tutorial -XIII** |
| **42-43** | Quadratic forms and nature of the quadratic forms |
| **44** | Summarization of Unit |
| | **Tutorial -XIV (**Short assessment) |
| **45-48** | **Revision** |

**Text Books:**

1. Erwin Kreyszig, *Advanced Engineering Mathematics*, 10th edition, John Wiley & Sons, 2011.
2. B. S. Grewal, *Higher Engineering Mathematics*, 45th edition, Khanna Publishers, 2020.

**Reference Books:**

1. Greenberg M D, *Advanced Engineering Mathematics*, 2nd edition, Pearson Education, Singapore, Indian Print, 2003.
2. Peter V. O'Neil, *Advanced Engineering Mathematics*, 7th edition, Cengage Learning, 2011.